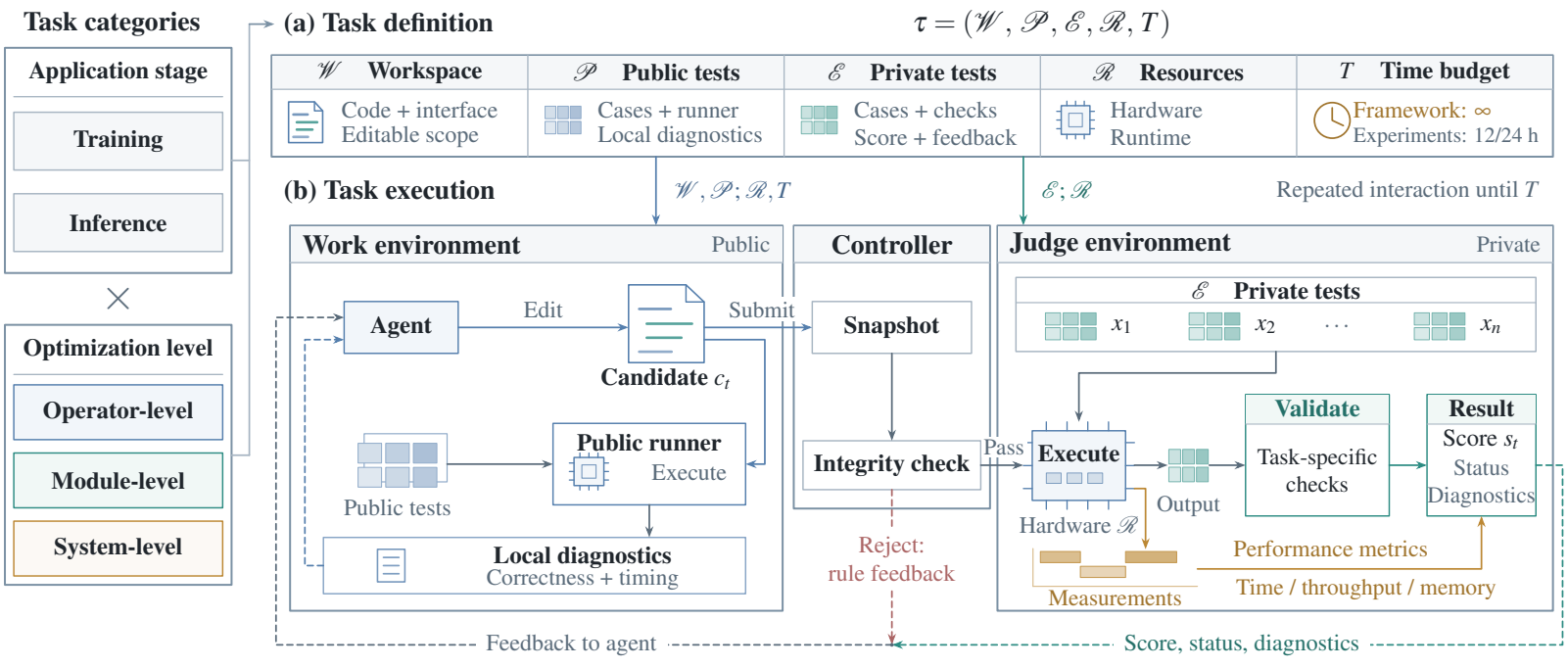
Task categories
Application stage
Training
Inference
×
Optimization level
Operator-level
Module-level
System-level
(a) Task definition
$\tau = (\mathscr{W}, \mathscr{P}, \mathscr{E}, \mathscr{R}, T)$
$\mathscr{W}$ Workspace
Code + interface
Editable scope
$\mathscr{P}$ Public tests
Cases + runner
Local diagnostics
$\mathscr{E}$ Private tests
Cases + checks
Score + feedback
$\mathscr{R}$ Resources
Hardware
Runtime
T Time budget
Framework: ∞
Experiments: 12/24 h
(b) Task execution
$\mathscr{W}, \mathscr{P}; \mathscr{R}, T$
$\mathscr{E}; \mathscr{R}$
Repeated interaction until T
Work environment
Public
Agent
Edit
Candidate c_t
Submit
Public tests
Public runner
Execute
Local diagnostics
Correctness + timing
Controller
Snapshot
Integrity check
Reject:
rule feedback
Judge environment
Private
$\mathscr{E}$ Private tests
x_1 x_2 ⋯ x_n
Pass
Execute
Hardware $\mathscr{R}$
Output
Validate
Task-specific checks
Result
Score s_t
Status
Diagnostics
Measurements
Performance metrics
Time / throughput / memory
Feedback to agent
Score, status, diagnostics